0.1 Standard Model of Particle Physics

The Standard Model (SM) of particle physics is a framework which contains our current understanding of the matter and three of the four four forces found in the universe. Within the SM there are 12 elementary fermions, 5 vector bosons which mediate the three fundamental forces and a scalar boson responsible for particle masses (Figure 1). It is important to note that the SM is not a complete model of the universe. A great example of this is the gravitational force which is not described by the theory, however at the small scales of of particle physics this force is neglected due to it's insignificant strength when compared to the other three forces (strong, electromagnetic and weak).

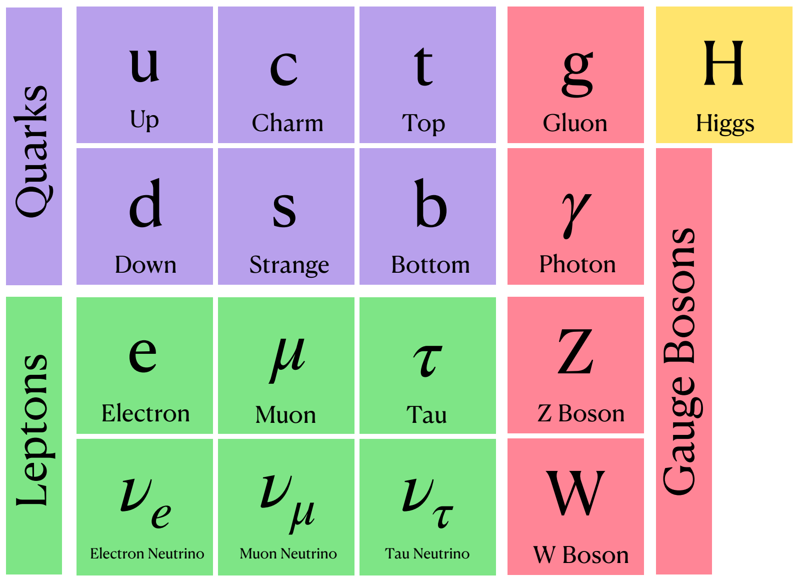

Figure 1: The Standard Model of particle physics with the 12 fermions shown in purple and green, the five vector bosons shown in red and the Higgs scaler boson shown in yellow.

The SM comprises of a quantum field theory (QFT) and is based on the non-Abelian gauge symmetry groups SU(3)×SU(2)×U(1). The SU(3) governs the strong interactions and the SU(2)×U(1) the electroweak interactions. Each group contains their respective gauge bosons: gluon for the strong and W, Z and photon for the electroweak. The SU(3) is an exact gauge symmetry where as the electroweak symmetry is spontaneously broken by the Higgs field, resulting in the massive electroweak bosons ($W^{\pm}$ and $Z^{\pm}$). Due to this broken symmetry which gives rise to massive quarks, the strong and weak eigenstates diverge. The weak eigenstates (d', s', b') of down, strange and bottom quarks which are associated to the weak interaction are related to the mass eigenstates (d, s, b) through the Cabibbo-Kobayashi-Maskawa (CKM) matrix V_{CKM} [1, 2].

$$\begin{bmatrix} d' \\ s' \\ b' \end{bmatrix} = V_{\mathrm{CKM}} \begin{bmatrix} d \\ s \\ b \end{bmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \begin{bmatrix} d \\ s \\ b \end{bmatrix} \tag{0.1}$$

Utilising the elements of the CKM matrix it is possible to calculate the probability that a quark type i will transition into type j with $|V_{ij}|^2$.

For a unitary 3x3 matrix the CKM can be reduced from nine parameters to four: three mixing angles (θ_{12}, θ_{13}, θ_{23}) which dictate the couplings between each generation of quarks and one phase (δ) which introduces CP violation [2]. A number of different parametrisations are possible but the standard choice is [3],

$$V_{\mathrm{CKM}} = \begin{bmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{bmatrix} \tag{0.2}$$

where $c_{ij} = \cos\theta_{ij}$ and $s_{ij} = \sin\theta_{ij}$. The mixing angles θ_{ij} can be selected to satisfy $0 \leq \theta_{ij} \leq \frac{\pi}{2}$ such that $c_{ij}, s_{ij} \geq 0$.

Through experiments it is known that there is hierarchical structure to the components of the CKM matrix,

$$s_{12} \ll s_{23} \ll s_{13} \tag{0.3}$$

and it is useful to further parametrise through the Wolfenstein parametrisation [4] (Equation 0.5). For this parametrisation the following definitions are used,

$$\begin{aligned} s_{12} &= \lambda \\ s_{23} &= A\lambda^2 \\ s_{13} &= A\lambda^3(\rho + i\eta) \end{aligned} \tag{0.4}$$

resulting in,

$$V_{\mathrm{CKM}} = \begin{bmatrix} 1 - \frac{\lambda^2}{2} & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \frac{\lambda^2}{2} & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{bmatrix} + O(\lambda^4). \tag{0.5}$$

Reducing the CKM to $O(\lambda^3)$ results in the loss of unity. However, unity can be restored by including $O(\lambda^5)$ [5] correction to the bottom left component of Equation 0.5,

$$V_{td} = A\lambda^3(1 - \bar{\rho} - i\bar{\eta}) \tag{0.6}$$

where

$$\begin{aligned} \bar{\rho} &= \rho(1 - \frac{\lambda^2}{2}) \\ \bar{\eta} &= \eta(1 - \frac{\lambda^2}{2}) \end{aligned} . \tag{0.7}$$

The unitary nature of the CKM matrix imposes conditions on the row and columns

$$\begin{aligned} \sum_i V_{ij} V^*_{ik} &= \delta_{jk} \\ \sum_j V_{ij} V^*_{kj} &= \delta_{ik} \end{aligned} \tag{0.8}$$

where δ_{jk} denotes the Kronecker delta function, which equals 1 when j = k and 0 otherwise. This results in six vanishing combinations of the CKM matrix which can be represented as triangles, with the most common being

$$V_{ud}V^*_{ub} + V_{cd}V^*_{cb} + V_{td}V^*_{tb} = 0 \tag{0.9}$$

and can be modified with the Wolfenstein parametrisation (Equation 0.10) which is depicted in Figure 2.

$$1 + \frac{V_{td}V^*_{tb}}{V_{cd}V^*_{cb}} = -\frac{V_{ud}V^*_{ub}}{V_{cd}V^*_{cb}} = \bar{\rho} + i\bar{\eta} \tag{0.10}$$

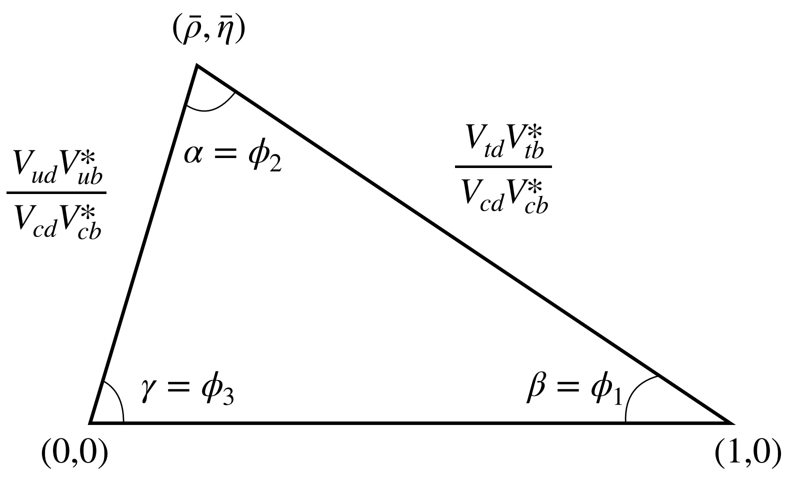

Figure 2: Sketch of the most common unitary triangle $V_{ud}V^*_{ub} + V_{cd}V^*_{cb} + V_{td}V^*_{tb} = 0$.

The CKM matrix components are fundamental parameters of the SM. Through a large number of experiments the elements of the CKM matrix can be measured and are combined to form a global fit [6]:

$$\begin{bmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{bmatrix} = \begin{bmatrix} 0.97436^{+0.00014}_{-0.00019} & 0.22498^{+0.00081}_{-0.00062} & 0.00373^{+0.00019}_{-0.00015} \\ 0.22484^{+0.00081}_{-0.00061} & 0.97351^{+0.00016}_{-0.00019} & 0.04160^{+0.00066}_{-0.00121} \\ 0.00857^{+0.00018}_{-0.00037} & 0.04088^{+0.00066}_{-0.00119} & 0.999125^{+0.000052}_{-0.000027} \end{bmatrix} \tag{0.11}$$

A feature of the CKM matrix is that the diagonal components which describe the transitions within the quark generations dominate, compared to the off-diagonal elements which are suppressed for transitions between one generation and doubly suppressed for transitions between two generations. This suppression between two generations produces rare decays, such as $K^+ \to \pi^+\nu\bar{\nu}$ described in Section .

Through measurement of the CKM matrix components it is possible to constrain the unitary triangle (Figure 2) shown in Figure

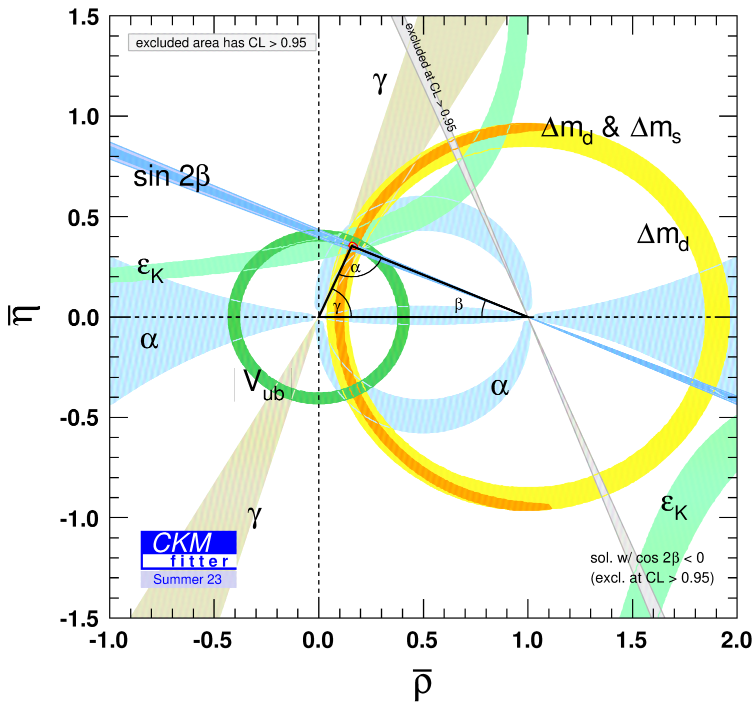

Figure 3: Constraints for the most common unitary triangle in the $(\bar{\rho},\bar{\eta})$ plane [6].

Including the four CKM components discussed above there are nineteen free parameters of the SM:

- Four CKM parameters discussed above.
- Six Quark masses (Up, down, charm, strange, top and bottom).
- Three Lepton masses (Electron, muon and tau).
- Three Gauge Couplings (U(1), SU(2) and SU(3))
- Two Higgs parameters (Vacuum expectation value and mass).
- Quantum chromodynamics (QCD) vacuum angle.

Each one of these parameters needs to be measured through experiment to the highest precision so that any discrepancies can be resolved. Any deviations from theoretical understanding could be hint of physics beyond the Standard Model (BSM).

0.2 Kaon Physics

The first possible observation a heavy meson was in 1943 by Leprince-Ringuet and L'heritier [7, 8] and predated even the discovery of the pion in 1947 [9]. Utilising a cloud chamber to measure cosmic rays they found the first evidence of a particle with a mass of 990 times the mass of the electron (0.511 MeV [10]). However, it was possible that the upper limit on the measured mass could reach the proton mass so they were not credited with the discovery. In 1947 while similarly studying cosmic rays with cloud chambers G. D. Rochester and C. C. Butler published a result which included two pictures showing forked tracks [11]. One of the tracks proved to show the decay of a neutral particle into two charged daughter particles, which turned out to be decay of the neutral kaon to two pions. The other track showed the decay of a charged particle into charged and neutral daughter particles. This result was later confirmed in 1950 by a group in Cal-Tech who observed multiple instances of these decays [12]. However, between these publications a group in Bristol which included physicist Rosemary Fowler, used emulsion chambers to observe the decay of a charged particle into three charged particles [13], know as the tau decay. This resulted in what was known as the tau-theta problem (names given to the respective particle decays), which eventually led to the discovery that parity is not conserved in weak interactions [14].

0.2.1 $K^+ \to \pi^+ \nu \bar{\nu}$

0.2.2 $K_L^0 \to \pi^0 l^+ l^-$

0.2.3 $K_L^0 \to \pi^0 \nu \bar{\nu}$

0.3 Dark Photon

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